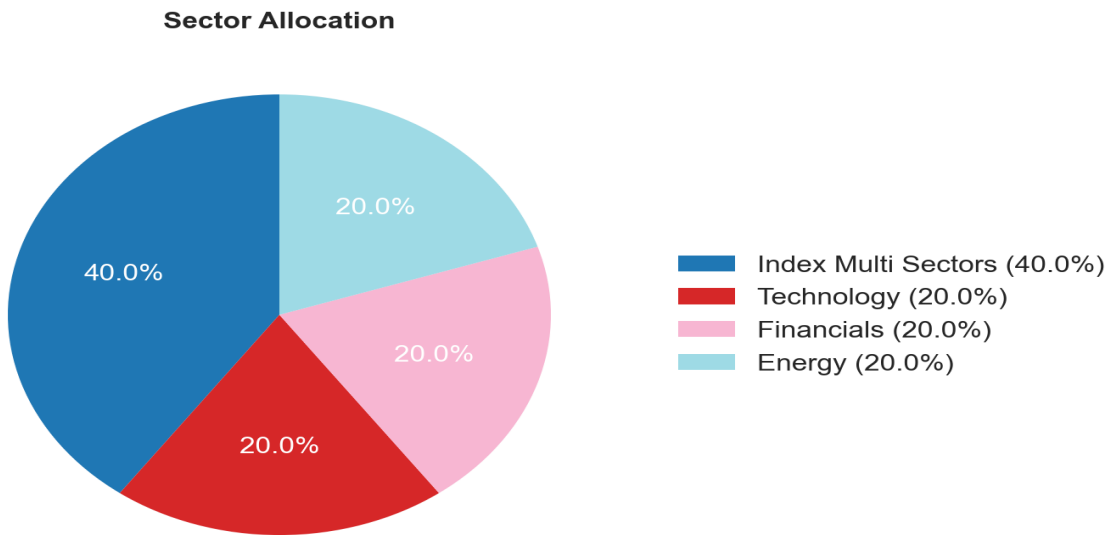


Portfolio rotation report

Date of extraction	2025-12-07 22:07
Data window	2020-01-02 → 2024-10-21
Frequency	Daily — 1209 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+504.5% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	QQQ, SPY, XLE, XLF, XLK
Indicators	Short term: 40 days volatility, 60 days return, 80 days momentum Long term: 100 days volatility, 120 days return, 240 days momentum

1. Portfolio Composition

- XLK (Technology Select Sector SPDR Fund — Technology)
 - XLF (Financial Select Sector SPDR Fund — Financials)
- XLE (Energy Select Sector SPDR Fund — Energy)
 - SPY (SPDR S&P; 500 ETF — Index Multi Sectors)
 - QQQ (Invesco QQQ Trust — Index Multi Sectors)



Macro Profile Insight

Macro coverage

- Partial coverage: several macro themes remain unrepresented.
- Broad balance across market capitalizations: diversification achieved across the size spectrum.

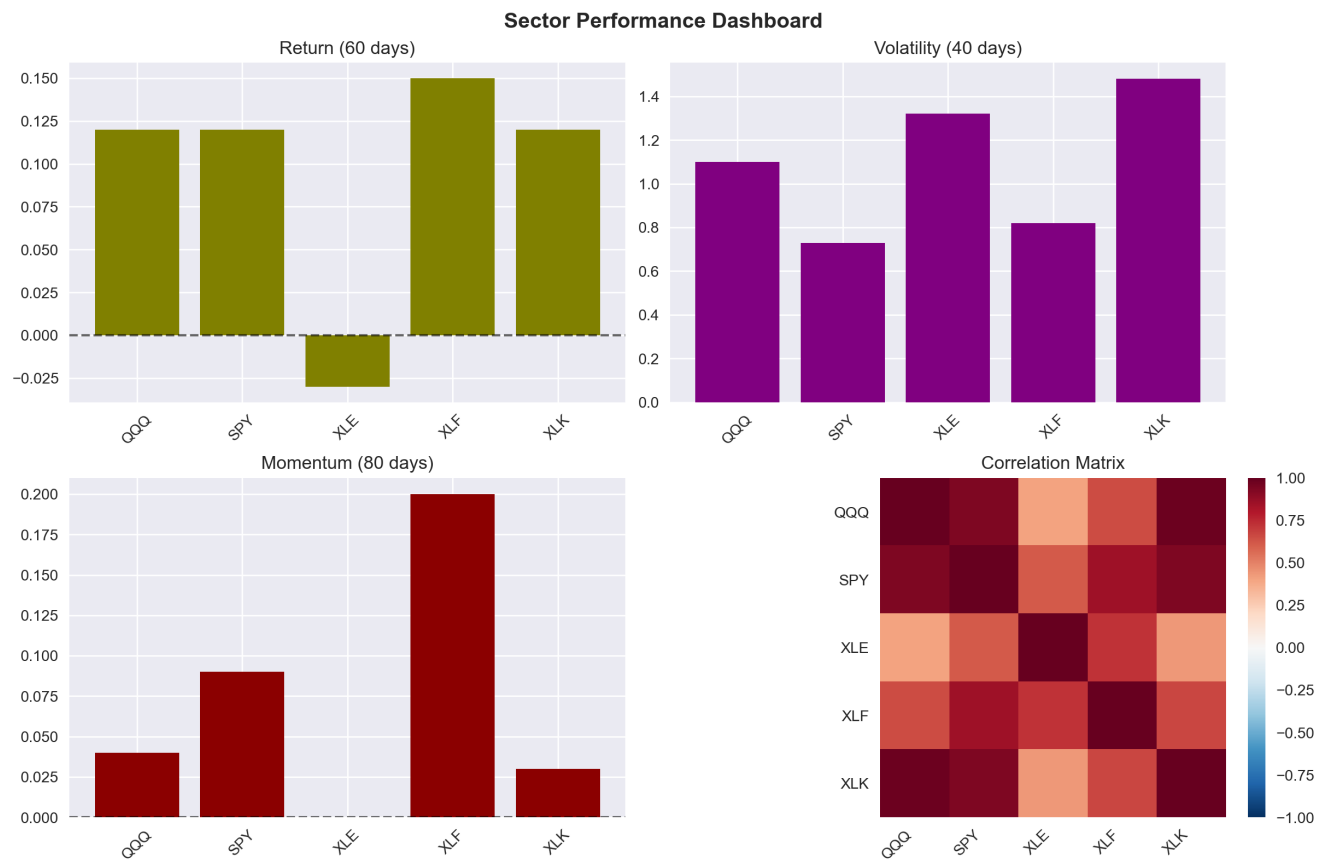
Structural properties

- Concentration risk: portfolio heavily exposed to etf instruments.
- Liquidity stance: pockets of thin/illiquid names; robust median dollar volume.
- Market-neutral beta exposure: close alignment with index dynamics.
- Volatility profile skewed to dynamic/speculative names.

Additional comments

- Significant sector gaps: broad portions of the economy remain uncovered.

2. Sector Performance Visualization



Top Positive Correlations

100.0% of portfolio

Pair	Correlation
QQQ / XLK	0.98
QQQ / SPY	0.93
SPY / XLK	0.93
SPY / XLF	0.85
XLE / XLF	0.72

Overall, average cross-sector correlation was 0.72, ranging from 0.40 to 0.98. This suggests very high correlation — sectors behave almost as a single macro block. Correlation dispersion remains meaningful, indicating both tightly and weakly linked sectors. Strongest co-movement between QQQ and XLK (0.98); weakest between QQQ and XLE (0.40). Market-driven regime: macro factors dominate individual sector dynamics.

3. Key Performance Indicators

Ticker	R (%)	V (%)	RAR	M (%)	Signal	Regime	Liquidity	Commentary
QQQ	0.12	1.10	0.105	0.04	neutral	subdued	liquid	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
SPY	0.12	0.73	0.165	0.09	neutral	subdued	liquid	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
XLE	-0.03	1.32	-0.019	0.00	neutral	subdued	liquid	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
XLF	0.15	0.82	0.18	0.20	neutral	subdued	thin	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, under limited liquidity.
XLK	0.12	1.48	0.081	0.03	neutral	normal	thin	mixed short-term dynamics, signal unclear or in transition, normal volatility regime, under limited liquidity.

Top by Average Return

Ticker	AvgReturn(%)
XLF	0.15
QQQ	0.12
SPY	0.12
XLK	0.12
XLE	-0.03

Top by Volatility

Ticker	Volatility(%)
XLK	1.48
XLE	1.32
QQQ	1.1
XLF	0.82
SPY	0.73

Top by Momentum

Ticker	Momentum(%)
XLF	0.2
SPY	0.09
QQQ	0.04
XLK	0.03
XLE	0.0

3.b Rotation & Stress Metrics

Ticker	ΔReturn (pp)	Momentum Ratio	Signal	Volatility Ratio	Beta	ADV10 USD	Interpretation
QQQ	-0.00	0.27	0.031	0.87	1.18	53.05 M	stable performance, momentum fading, volatility normalizing, amplifying market trends, under thin liquidity regime.
SPY	-0.01	0.60	0.056	0.86	1.00	72.91 M	stable performance, momentum fading, volatility normalizing, amplifying market trends, under thin liquidity regime.
XLE	0.07	0.00	0.0	1.06	0.80	28.61 M	stable performance, momentum fading, volatility normalizing, amplifying market trends, under thin liquidity regime.
XLF	0.02	1.13	0.094	0.93	0.84	36.84 M	stable performance, momentum stable, volatility normalizing, amplifying market trends, under thin liquidity regime.
XLK	-0.01	0.20	0.025	0.88	1.33	16.33 M	stable performance, momentum fading, volatility normalizing, amplifying market trends, under thin liquidity regime.

■ Light-impact observations

For **QQQ, SPY, XLE, XLF**: *Volatility compression* — subdued volatility; a breakout setup may be forming. (avg ann. return 0.1%, momentum 0.1%, vol 1.0%)
For **XLK**: *Range / noise* — no clear direction; rotating momentum and mid-range volatility. (avg ann. return 0.1%, momentum 0.0%, vol 1.5%)